DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**ACUTE MANAGEMENT OF SEIZURES**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
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1.0 PURPOSE

1.1. For safe and standardized management of seizure disorder

2.0 DEFINITION/ABBREVIATION

2.1. Seizure: defined as a sudden attack of altered behavior, consciousness, sensation or autonomic function produced by transient disruption of brain function.

2.2. Abbreviations:

2.2.1. Sz: Seizure

2.2.2. NS: normal saline

2.2.3. mL: milliliter

2.2.4. BGL: Blood glucose level

3.0 RESPONSIBILITY

3.1 **PICU consultant/specialist** in-charge of the unit responsible for the medical care of the patients

3.2 **Head Nurse:** Responsible in ensuring that all staffs are competent in handling procedure and in monitoring the actual performance of the staff nurse.

3.3 **Staff nurse:** Responsible in carrying out doctor's order and monitoring patient's condition.

4.0 POLICY

4.1. Proper assessment should be done.

4.2. Patient's possible complication should be identified.

4.3. Seizure precaution to be implemented.

4.4. Treatment according to Algorithm must be followed.

5.0 PROCEDURE (S)

5.1. Assessment and management- The first step in the management of the patient who is having a seizure is to assess and support airway, breathing and circulation. This will ensure that the seizure does not compromise supply of oxygenated blood to the brain and is not secondary to hypoxia and/or ischemia.

5.1.1. Airway- A clear airway is the first requisite. If the airway is not clear it should be opened and maintained with a head tilt/chin lift or jaw thrust maneuver while the child is in a supine position. An oropharyngeal or nasopharyngeal airway may be used. will generally control the airway. Even if the airway is clear, the oropharynx may need secretion clearance by gentle suction. After initial airway clearance the airway should continue to

be observed and protected as required. Post seizure a child should be positioned on his or her side (recovery position).

5.1.2. Breathing- Assess the following for adequacy of breathing by the 'look, listen and feel' method:

5.1.2.1. Effort of breathing:

5.1.2.1.1. Recession

5.1.2.1.2. respiratory rate

5.1.2.1.3. grunting, this may be caused by the convulsion and not be sign of respiratory distress in this instance.

5.1.2.2. Efficacy of breathing:

5.1.2.2.1. breath sounds

5.1.2.2.2. chest expansion/abdominal excursion

5.1.2.2.3. monitor oxygen saturation with a pulse oximeter.

5.1.2.3. Effects of breathing

5.1.2.3.1. heart rate

5.1.2.3.2. skin color

5.1.2.4. All fitting children should receive high flow oxygen through a face mask with a reservoir as soon as the airway has been demonstrated to be adequate.

5.1.2.5. Prolonged seizures and/or repeated doses of anti-epileptic medications may lead to compromise of airway and breathing requiring ongoing support including intubation. Help from senior clinicians, should be obtained for intubation.

5.1.3. Circulation- Assess the following for adequacy of cardiovascular status:

5.1.3.1. Heart rate: the presence of an inappropriate bradycardia or hypertension will suggest raised intracranial pressure.

5.1.3.2. Pulse volume: assess the adequacy of circulation by palpation of central pulse (femoral, brachial)

5.1.3.3. Capillary refill: capillary refill should be three seconds or less and is measured by applying coetaneous pressure on the center of the sternum for 5 seconds

5.1.3.4. Blood pressure: significant (>97th percentile for age) hypertension indicates a possible etiology for the seizure.

5.1.3.5. Effects of circulatory inadequacy on other organs pale, cyanosed or cold skin. Gain intravenous access. If vascular access is not readily obtained, initial doses of antiepileptic should be given by the buccal, intra-nasal or intramuscular route. Intraosseous access should be obtained immediately in children with signs of shock if intravenous access is not readily obtained. Intraosseous access may be needed for administration of long-acting anti epileptics if there is no intravenous access after two doses of a benzodiazepine.

5.1.3.6. Give 20 *mL/kg* rapid bolus of 0.9% sodium chloride to any patient with signs shock or sepsis

5.1.4. Disability- Assess the following for adequacy of neurological function:

5.1.4.1. The AVPU (Alert, Voice, Pain, Unresponsive) score cannot be measured meaningfully during a seizure as a generalized seizure depresses the level of consciousness.

5.1.4.2. Pupillary size, reaction and symmetry should be noted. Pupillary changes can occur during a seizure but may also result from poisoning or raised intra-cranial pressure. Very small pupils suggest brainstem injury or opiate poisoning, large pupils suggest amphetamines, atropine, or tricyclic antidepressants.

5.1.4.3. Note the child's posture. Decorticate or decerebrate posturing in a previously normal child should suggest raised intracranial pressure. These postures can sometimes be mistaken for the tonic phase of a seizure. Consider also the possibility of a drug induced dystonia that is distinguishable from tonic-clonic status epilepticus.

5.1.4.4. Assess for neck stiffness in a child and a bulging fontanelle in an infant, which suggests meningitis.

5.1.5. Exposure

5.1.5.1. Look for rash and bruising as signs of sepsis or injury

5.1.5.2. Measure temperature

5.1.6. Fluids

5.1.6.1. Correct any fluid or electrolyte imbalance according to established protocols.

5.1.7. Glucose

5.1.7.1. Don't Ever Forget Glucose

5.1.7.2. Take blood glucose stick test and laboratory test.

5.1.7.3. Give 2 mL/kg of 10 percent glucose to any hypoglycemic patient (BGL «3mmolIL).

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5.1.7.4. If possible, take 10 mLs of clotted blood before giving the glucose for later investigation of the hypoglycemic state. Ongoing monitoring and reassessment of A-G

5.1.7.5. After each dose of anti-epileptic medication

5.1.7.6. Every 5 minutes while the seizure continues

5.1.7.7. Every 15 minutes after a seizure until level of consciousness returns to normal

6.0 EQUIPMENTS

6.1. Glucometer

6.2. Gluco strip

6.3. Airways (oral and nasopharyngeal)

7.0 ATTACHMENT/FORMS

7.1. List of Medications for seizures

8.0 REFERENCES

8.1. King Fahad Hospital, Policy and Procedure for Management of Hypoglycemia in Children, Policy Number: MOHIC-PICU 4/3 (4)

9.0 REVISION CLAUSE

9.1. N/A